

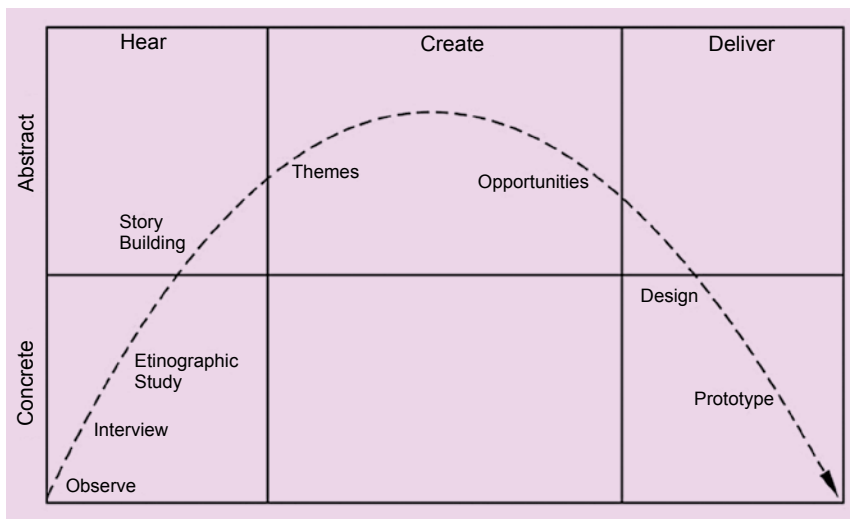


Fig. 4: IDEO HCD design steps

to adopt a human-centred approach to problem-solving and innovation. Their emphasis on empathy, collaboration, and iterative design processes continues to shape the field of design and influence how businesses approach innovation.

According to IDEO philosophy, a successful design must combine people's needs, technological feasibility, and business sense. It should leverage the opportunities offered by technological progress and innovation. People are at the centre of IDEO's design processes.

According to the IDEO process, the design must address four pleasures experienced by people. These four pleasures are:

Physical pleasure. Appeals to the five senses—sight, hearing, smell, touch, and taste.

Social pleasure. Esteem value. The product should make people proud of owning it. Emphasising a product as cheap and meant for the poor diminishes its esteem value. Good looks, brands, appealing logos, feel-good advertisements, and socially responsible products address this pleasure.

Psychological pleasure. Owning the product should give a sense of achievement. This is why people

often crave premium products when a cheaper alternative would suffice their needs.

Ideological pleasure. Arises from doing the right thing. It may stem from factors like environmental friendliness and sustainability, or a sense of belonging, such as 'Made in India.'

People have three types of emotional responses to these four pleasures. The emotional responses are visceral, behavioural, and reflective.

The visceral level is responsible for immediate emotional responses, which come from deeply ingrained, almost animalistic qualities of human emotion. These responses are almost entirely out of our control and heavily influenced by cultural heritage. For example, we associate gold with costly items, while colours like brown and blue give a formal look, and red stands out as bold and flashy.

The behavioural level refers to controlled aspects of human action, where we unconsciously analyse a situation to develop goal-directed strategies likely to be effective in the shortest time or with the fewest actions possible. These emotions are prominent during product use. Simple products that are frequently used reinforce habit formation.

The reflective level, as described by Don Norman, is "the home of reflection, conscious thought, learning new concepts, and generalisations about the world." While these three levels are classified as separate dimensions of the emotional system, they are linked and influence one another, contributing to our overall emotional experience of the world.

The IDEO design (HCD) process consists of three main phases:

Hear. Collect stories and inspiration from the people you are designing for. Decide where to gather information and plan your market research accordingly. Document your observations and create use case stories to generate different situations where the product will be used. From these observations, derive a list of requirements.

Create. Design to translate the requirements into reality. Move from concrete to abstract thinking by identifying themes and opportunities, and then transition back to the concrete with solutions and prototypes.

Deliver. Begin realising the solutions by refining your top solutions and moving them towards implementation through revenue and cost modelling, capability assessment, and implementation planning.

Technical requirements

In addition to styling and ease of use, we must address basic technical requirements. Some of these requirements are mandatory and specified by governmental agencies, while others are driven by quality standards.

Environmental protection. Standards like NEMA (National Electrical Manufacturers Association), IP66-67, and others are widely accepted norms that provide a certain level of protection against environmental hazards. These standards ensure a common understanding and interoperability.